

inverter (with power conversion efficiency of, say, 80 per cent), you would be able to use the appliance for an aggregate of 7.5 hours ($40W \times 7.5 \text{ hours} \times 100/80 = 375 \text{ watt-hours}$).

As per the recommendations of

International Energy Agency, one should draw not more than 40 per cent of the rated energy capacity (VAh) of a battery during a 24-hour period. Accordingly, your battery should be rated for at least $375 \times 100/40$

$= 937.5 \text{ watt-hours}$ of energy. With the battery terminal voltage of 12V, its capacity should be around 80 Ah (energy capacity $= 12V \times 80 \text{ Ah} = 960 \text{ Ah}$). Tubular-type battery is the preferred choice for a longer life as this application needs batteries having deep-discharge capability.

Wire sizing. The voltage drop between PV modules and batteries may not be allowed to exceed 2 per cent of the 12V battery terminal voltage. For $5A I_{MPP}$ the maximum length (in metres) of various two-conductor copper wires is given below:

SWG#	16	14	#12	#10	#8	#4	#2
Metre	2.5	4	6	10	15	24	40

For other PV module amperage, you may refer the chart at www.solar4power.com/solar-power-volt_drop.html#Volt%20Drop%20Chart

Important terms. Important

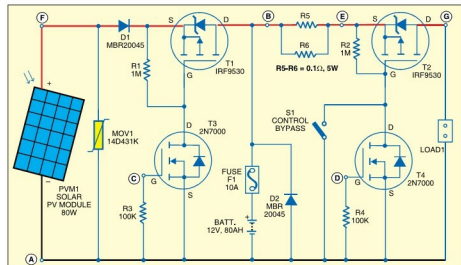


Fig. 5: Main charger circuit

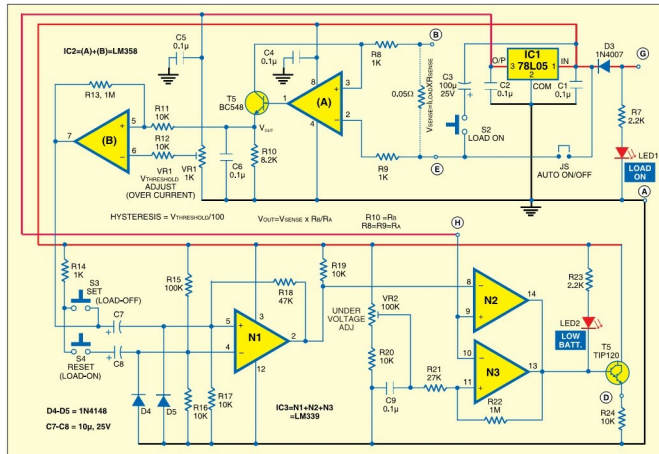


Fig. 6: Over-load cum low-battery control section